

FinnPRIO: a model for ranking invasive plant pests based on risk

Jaakko Heikkilä · Juha Tuomola ·
Eija Pouta · Salla Hannunen

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Abstract The number of plant pests that may be transported to new areas with international trade is too great for subjecting all of them to a full scale pest risk assessment. There is therefore a need for a quick risk assessment procedure that also ranks the pests according to their risk. The FinnPRIO model can be used to assess the risk of alien plant pests for Finland. It follows the basic structure of a full scale pest risk assessment, i.e. it can be used to separately estimate the probabilities of entry, establishment (including spread), and the likely impacts. The model also includes a section for assessing preventability and controllability of a pest invasion. The model consists of multiple-choice questions with answer options yielding a different number of points. For each question the most likely answer option and the plausible minimum and maximum options are chosen. The total risk score is simulated using a PERT distribution, providing a scale of potential risk for each pest and indicating the level of uncertainty associated with the assessment. The model is accompanied by a guide for the interpretation of the

questions and answer options. The model's functionality has been tested through simulations, and it has been validated by comparing pest rankings produced using the model to those obtained in expert workshops. To date, 95 pests have been assessed with the model. The results indicate that the model is well capable of differentiating pests based on their estimated risk.

Keywords Invasive alien species · Plant health · Prioritisation · Ranking · Risk assessment · Expert judgment

Introduction

Growth in international trade and travel together with changes in climate and production practices have significantly increased the rate of plant pest introductions to new areas (Santini et al. 2013; Dalmazzone and Giaccaria 2014). Restriction of international trade to prevent the pest introductions is acceptable under the terms of the World Trade Organization Agreement on Sanitary and Phytosanitary Measures as long as the restrictions are based on risk assessment that has been carried out in accordance with the relevant International Standards for Phytosanitary Measures (ISPM standards) (IPPC 2007, 2013).

All risks cannot be assessed a priori, since all the species that may become pests if transported to new areas are not known. However, the number of known potentially invasive pest species alone is so great that

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J. Heikkilä (✉) · E. Pouta
Natural Resources Institute Finland, Helsinki, Finland
e-mail: jaakko.heikkila@iki.fi

J. Tuomola · S. Hannunen
Finnish Food Safety Authority Evira, Helsinki, Finland

they cannot all be subjected to a full-scale pest risk assessment and cannot all be thoroughly monitored by plant health officials. Priority species need to be selected for full scale risk assessment, and plant health inspections and stakeholder information should be targeted to deal with the most important pests. There is therefore a need for a risk assessment procedure that is lighter and faster than the standard full scale pest risk assessment and that also ranks the pests according to the risk that they present.

Scoring models provide an answer to this need. A scoring model is here defined as a structured system that places pests into a ranking order by asking the assessor a series of questions. The answers to individual questions are scored and the scores are aggregated to form a total risk score, which can be used to rank the species. The rank of the species denotes the risk that the species presents to the society.

The greatest advantage of scoring models over unstructured expert judgement is that the models facilitate consistency and transparency of assessments. Compared to full-scale risk assessments the results of scoring models are often more uncertain and due to their basic nature, all species-specific factors cannot be fully taken into account. Nonetheless, the use of such scoring models in policy support is relatively high. In a comprehensive review, Leung et al. (2012) found that more than half (57 %) of the alien species scoring models had been used for policy support, whereas for only 6 % of the models it was confirmed that they had not been used for policy support. For instance, in Australia a scoring model, initiated in 1997, had in the first 8 years of operation been applied to plant imports of over 2800 plant species (Weber et al. 2009).

Heikkilä (2011) concluded that there seems to be a fair amount of alien species scoring models available for practical application, especially in the case of invasive plants. However, only a select few models have become widely used, possibly due to the clarity and performance of the models as well as their general applicability. The most widely applied scoring models are the Australian Weed Risk Assessment (WRA) (Pheloung et al. 1999) and the Reichard-Hamilton decision tree (Reichard and Hamilton 1997). The Australian WRA has been found to perform well in different regions and even for different taxa (Leung et al. 2012). In the past couple of years, several new models for ranking invasive species, including plant

pests, have been developed, particularly in Europe. These include the UK Plant Health Risk Register (Baker et al. 2014), the Belgian *Harmonia*⁺ and *Pandora*⁺ (D'hondt et al. 2015), and the French model (Moignot and Reynaud 2013). There are also recent scoring models that concentrate on assessing the impacts of invasive species, including the Unified Classification system (Blackburn et al. 2014b; Hawkins et al. 2015), the German-Austrian black list GABLIS (Essl et al. 2011) and the Norwegian impact assessment (Sandvik et al. 2013). Most of the recent developments are compatible with the very comprehensive European and Mediterranean Plant Protection Organization (EPPO) risk assessment protocol (EPPO 2011).

Despite the recent developments of pest risk scoring models, none of the models has emerged as the standard risk ranking method for invasive plant pests. This may be partly due to slight differences in the applications for which the models have been developed (import assessment, impact assessment, post-border management assessment, and so on). Yet, it also indicates that there still is a need to explore the scoring approaches further.

Recent reviews (Heikkilä 2011; Leung et al. 2012) have made a number of suggestions on how risk scoring models for alien species should be structured. For example they recommend that all the main components of risk, i.e. probability of entry, establishment, spread and the likely impacts, should be included in the model, and that the total risk score should be formed by multiplying the aforementioned components with each other. They also advise to include an assessment of the feasibility of management and an assessment of all potential impact categories, i.e. economic, social and environmental, in the model. Accounting for uncertainty of the assessment is considered especially important.

This paper presents the FinnPRIO pest prioritisation model that enables comparison of all types of plant pests and takes into consideration the critical recommendations made in the recent reviews (Heikkilä 2011; Leung et al. 2012). The aims in this paper are threefold: (1) to present the structure and validation of the FinnPRIO model; (2) to demonstrate its functionality by using it to score 95 plant pests; and (3) to discuss the challenges in developing invasive pest scoring models and in presenting their results for policy recommendations.

The FinnPRIO model

Basic structure

The FinnPRIO model follows the basic structure of a full scale pest risk assessment, i.e. it has separate sections for assessing the probabilities of entry, establishment and spread, and the likely impacts (IPPC 2007, 2013, see also Leung et al. 2012 for the TEASI framework). However, since differentiating between establishment and spread in a scoring model is not clear-cut (see Hayes and Barry 2008 for a review), in the FinnPRIO model they are combined into a single section called establishment. Additionally, there is a section where the manageability of pest entry, establishment and spread are assessed.

The model consists of discrete multiple-choice questions with different answer options yielding a different number of points (Table 1). Each question has 3–5 answer options, except for one question that has 13 options and two questions that employ binary yes/no answers. The questions in the model were derived using the international standard for pest risk analysis, ISPM 11 (IPPC 2013), as the general framework. In addition, a list of about 800 questions derived from existing scoring models was consulted.

The scores for the sections, i.e. entry, establishment, impact, and management, are calculated with specifically designed formulae (presented later). A score describing the likelihood of invasion is produced as the product of entry and establishment scores. The section scores and the likelihood score are normalised such that they all have a minimum attainable level of 0 and a maximum attainable level of 100. The total risk score of the FinnPRIO model is calculated as the product of the section scores, i.e. entry, establishment, and impact sections. The total score is scaled to range from 0 to 1000.

A guide has been developed for assisting in the answering of the questions of the FinnPRIO model (available as supplementary material). The guide provides instructions on which factors should be considered and which should be ignored when answering the questions. Also detailed guidance on the use of the different answer options is given.

Entry

The entry section consists of eight potential transport pathways (A–H), of which the assessor can choose the

ones considered possible for the assessed pest. Pathways A–E deal with different types of host plant material i.e. seeds (A), plants for planting (B), wood and wood products (C), food and fodder (D), and cut flowers and branches (E). Pathway F deals with commodities other than host plant material, as well as transport media and passengers ('hitchhiking'). Pathway G deals with natural spread, and pathway H with intentional introduction, for instance for biological control purposes.

The section consists of two questions common to all pathways and two pathway-specific questions (Table 1). The question related to the geographical distribution of the pest (ENT1) is taken into account for all other pathways except for natural spread and intentional introduction (pathways G–H). The question dealing with the potential of the pest to be transported through the pathway (ENT2) is common to all pathways. ENT2A and ENT2B are alternative questions that differ only in whether the current entry risk management measures are ignored (ENT2A) or taken into account (ENT2B) when answering the question. The first option is appropriate when considering deregulation of a pest, and the latter when considering the need for further regulation. By comparing the two it is possible to assess the effect of the present regulations on the risk. The question related to the volume of trade (ENT3) is taken into account only for the pathways dealing with host plant material (pathways A–E), and not for hitchhiking, natural spread or intentional introduction (pathways F–H). Finally, the likelihood of the pest reaching a suitable habitat in Finland through the pathway (ENT4) is considered for all the pathways.

The scores for the question dealing with the volume of trade (ENT3) are conditioned on the answer option chosen for the question about the potential of the pest to be transported through the pathway (ENT2A/B). If transport through the pathway is considered likely or very likely, the volume of trade plays a lesser role in determining the total entry risk, as even a very low number of incoming consignments could be sufficient for the pest entry (Table 2).

The score for each pathway is the product of its question scores calculated either using ENT2A (without current entry risk management measures) or ENT2B (with current entry risk management measures), and converted to probabilities such that the maximum score, i.e. probability of 1, is the same for

Table 1 The questions and answer options of the FinnPRIO model

Question	Answer options
ENT1: How wide is the current global geographical distribution of the pest? (pathways A–F)	a. Small (1) b. Medium (2) c. Large (3)
ENT2A: Not taking into account current management measures, can the pest Be transported in international trade with the host plant commodity considered in the pathway (pathways A–E)? Be transported from one country to another with other than host plant commodity, transport or passengers (pathway F)? Spread naturally to Finland from its current ranges during the next ten years (pathway G)? Be intentionally introduced to Finland (pathway H)?	a. No it cannot (0) b. It can, but it is very unlikely (0.5) c. It can, but it is unlikely (1) d. It can, and it is likely (2) e. It can, and it is very likely (3)
ENT2B: As in ENT2A, but taking into account current official entry management measures	As in ENT2A
ENT3: How large a volume of the considered host plant commodity is traded into Finland annually (pathways A–E)?	a. Non-existent (see Table 2) b. Small (see Table 2) c. Medium (see Table 2) d. Large (see Table 2)
ENT4: Can the pest transfer to a suitable habitat after entering Finland via the pathway?	a. It cannot (0) b. It can, but it is very unlikely (0.5) c. It can, but it is unlikely (1) d. It can, and it is likely (2) e. It can, and it is very likely (3)
EST1: Could the pest reproduce and overwinter in Finland taking into account the climate and production conditions?	a. No it could not (0) b. It could, but it is unlikely (1.5) c. It could, and it is likely (4.5) d. It could, and it is very likely (9)
EST2: In how large an area do the pest's host plants grow or are cultivated in Finland?	a. Not at all (see Table 3) b. Very small (see Table 3) c. Small (see Table 3) d. Medium (see Table 3) e. Large (see Table 3)
EST3: How quickly would the pest likely spread in Finland?	a. Very slowly (see Table 3) b. Rather slowly (see Table 3) c. Rather quickly (see Table 3) d. Quickly (see Table 3)
EST4: Does the pest have characteristics that could assist in its establishment or spread in new areas?	a. No it does not (0) b. It has characteristics that could assist to some extent (1) c. It has characteristics that could assist to a great extent (2) d. It has characteristics that could assist to a very great extent (3)

Table 1 continued

Question	Answer options
IMP1: How significant are the direct economic losses that the pest would cause in Finland?	a. It would not cause losses in Finland (0) b. <0.05 million € per year (0.5) c. 0.05–0.1 million € per year (1) d. 0.1–0.2 million € per year (1.5) e. 0.2–0.4 million € per year (2) f. 0.4–0.8 million € per year (2.5) g. 0.8–1.5 million € per year (3) h. 1.5–3 million € per year (3.5) i. 3–6 million € per year (4) j. 6–12 million € per year (4.5) k. 12–25 million € per year (5) l. 25–50 million € per year (5.5) m. >50 million € per year (6)
IMP2: Would the pest cause the following indirect economic impacts in Finland? Would the pest impact foreign trade? Is the pest a vector for other pests? Would the pest have a significant impact on the profitability of some plant production sector?	a. No (0 for each question) b. Yes (1 for each question)
IMP3: How much direct impact would the pest have on the natural ecosystems in Finland?	a. No impact (0) b. Moderate impact (2) c. Significant impact (4) d. Very significant impact (6)
IMP4: Would the pest have the following environmental or social impacts in Finland? Cultural impacts Significant aesthetic impacts An impact on plants which have an important, recognized position in the Finnish culture	a. No (0 for each question) b. Yes (1 for each question)
MAN1: Can the pest spread naturally to Finland from its current range during the next ten years?	a. No it cannot (0) b. It can, but it is unlikely or very unlikely (2) c. It can, and it is likely or very likely (4)
MAN2: Is the pest present in the area of the European Union?	a. No it is not (0) b. Yes in a small area (2) c. Yes in a large area (3)
MAN3: How difficult is it to detect the pest during inspections?	a. Easy (0) b. Difficult (1) c. Nearly impossible (2)
MAN4: How difficult would it be to eradicate the pest from Finland?	a. Easy (0) b. Rather difficult (2) c. Very difficult (3) d. Impossible (4)

Table 1 continued

Question	Answer options
MAN5: How difficult would it be to survey the pest's occurrence in Finland?	a. Easy (0) b. Rather difficult (1) c. Very difficult (2) d. Impossible (3)

If the question is not considered for all the pathways the relevant pathways are indicated in brackets. The score associated with the different answer options are shown in brackets

Table 2 The matrix used in conditioning the score of ENT3 (volume of trade) on the answer option chosen for ENT2 (transport potential)

ENT3	ENT2				
	a (no)	b (very unlikely)	c (unlikely)	d (likely)	e (very likely)
a (non-existent)	0	0	0	0	0
b (small)	0	1	1	1	3
c (medium)	0	2	2	3	3
d (large)	0	3	3	3	3

Table 3 The matrix used in creating SPR1 (spread parameter) based on answer options chosen for EST2 (host plant area) and EST3 (rate of spread)

EST3	EST2				
	a (not at all)	b (very small)	c (small)	d (medium)	e (large)
a (very slowly)	0	3	2	1	1
b (rather slowly)	0	5	4	3	2
c (rather quickly)	0	7	6	5	4
d (very quickly)	0	9	8	7	6

all pathways regardless of the number of questions. The total score for the entry section is obtained from the individual pathway probabilities through joint probability calculation. The probability for each pathway is considered independent, but not mutually exclusive from the probabilities of the other pathways.

Establishment

In the FinnPRIO model establishment and spread are assessed in the same section. The section consists of four questions (Table 1). The questions relate to the suitability of climate and production conditions (EST1), the host plant area (EST2), the rate of spread of the pest in Finland (EST3), and the pest's

characteristics that promote its establishment or spread in new areas (EST4).

The answer options chosen for the host plant area (EST2) and the rate of spread (EST3) are combined to get a score for a specific spread parameter (SPR1) as presented in Table 3. The spread parameter (SPR1) measures the relative time that different pests are expected to need to spread throughout the host plant area. The maximum spread scores therefore accrue to pests with very small host plant area (EST2) and a fast spread rate (EST3). To give similar weight to the questions related to establishment and spread, the maximum scores of the spread parameter (SPR1) and the suitability of climate and production conditions (EST1) were equalized.

The total score for the establishment section is the sum of the question scores (EST1 and EST4) and the spread potential (SPR1). However, if the climate and production conditions in Finland are not suitable for the pest (EST1 = a) or if host plants are not present in Finland (EST2 = a), the score for the whole section is zero.

Impact

The magnitude of the likely impacts of the pest is assessed with four questions (Table 1). The first two questions (IMP1 and IMP2) deal with economic impacts, and the latter two questions with environmental (IMP3) and social impacts (IMP4). For both economic and environmental impacts, the first question (IMP1 and IMP3) asks for an estimate of the magnitude of the pest's direct impacts in case it would spread to its entire suitable habitat in Finland. The other questions (IMP2 and IMP4) both include three separate questions with yes/no answers that capture other potential economic, environmental and social impacts of the pests.

The total impact score is calculated as a sum of the question scores such that different weights can be given to economic (IMP1 and IMP2) and environmental/social impacts (IMP3 and IMP4). The used weights should reflect the values of the appropriate target group, e.g. decision makers or the public. However, in assessments that need to be compared with each other, similar weighing should be used.

Management

In the management section the preventability of pest entry and the controllability of the pest after entry are assessed in separate subsections (Table 1). The preventability subsection has three questions which deal with natural spread potential of the pest (MAN1), presence of the pest in the European Union (MAN2), and detectability of the pest in inspections (MAN3). The controllability subsection has two questions which deal with feasibility of eradication (MAN4) and surveillance (MAN5).

The maximum question score within each subsection determines the total score for that subsection. The scores for preventability and controllability can be presented separately, or they can be combined into a general manageability score, by taking the minimum

from the subsection scores. A pest invasion can be managed either by prevention or by reactive control, and therefore the minimum score from the subsections can be used to determine the overall manageability score.

Uncertainty

In the FinnPRIO model uncertainty is considered with respect to all the questions. The questions are answered using the most likely answer option and additionally with the minimum and maximum plausible answer options. These are subsequently used to define a PERT probability distribution that describes the uncertainty of the answer. The PERT distribution is a modified Beta distribution that is commonly used for modelling expert opinion data (van Hauwermeiren and Vose 2009). The probability distributions of the different questions are then used to carry out a Monte Carlo simulation to obtain the probability distribution of the section scores, subsequent total risk scores and the final ranking of the pests. The probability distribution of the section scores and the total risk scores indicate the level of uncertainty inherent in the scores.

Ranking

The results of the FinnPRIO model can be used to rank pests according to their entry, establishment, likelihood, impact, or total risk score. The ranking can be done based on various parameters describing the probability distribution of the section scores and the total risk scores. Maximum values or upper percentiles are appropriate if a precautionary approach is used, whereas minimum or lower percentiles may be used for a laissez-faire approach. Also the mean or median of the score distribution may be used. However, since the probability distributions of the scores can be of any shape, the mode of the distribution may also be a suitable parameter, if the aim is to mimic the way in which an expert would summarise a visual probability distribution into a single most likely risk figure. Rankings that consider the entire probability distributions can be obtained using the concepts of stochastic dominance or mean–variance frontiers (Yemshanov et al. 2013a), although the latter can be applied only to normally distributed data. The option demonstrated in this paper is to use the whole probability distribution to get the rankings with a Monte Carlo simulation. The

total risk scores or the section scores can be used to rank the pests separately in each iteration of the simulation. It is then possible to draw summary statistics based on the rank of the pest rather than on the risk score of the pest. The advantage of this approach is that it reveals how volatile or robust the model is regarding the order in which the pests are ranked. The simulated rank distributions can also be used to assess, for instance, the probability that a pest is ranking higher than another pest.

Validation of the FinnPRIO model

Exploring the behaviour of the model

The theoretical ability of the FinnPRIO model to differentiate between pests, and the relative importance of different questions for their corresponding section scores and total risk score were studied by simulation with random numbers. Two types of randomisations were conducted: (1) the answers to all questions were randomised, and (2) the answer to each question was set at its minimum and maximum value, one at a time, while all other responses were randomised. In both cases it was assumed that host plants occur in the country ($EST2 \neq a$), and equal weight was given to economic (50 %) and environmental/social impacts (50 %). 1500 iterations were run for each case, and the resulting means of the section scores and total risk scores were recorded.

The results of the first random simulation (i.e. where the answers to all questions, except for $EST2$, were randomised) were used to evaluate the theoretical ability of the model to differentiate between the pests. The relative importance of the different questions on the section scores and the total risk score was then studied by comparing these results to the results of the second randomisation, where one question at a time was set at its minimum and maximum values. Such random simulation analyses were carried out several times when developing the model, and the results were used in fine-tuning the questions, answer options and formulae for calculating the section scores.

The random simulations carried out with the final model indicated that the model is in theory capable of differentiating between the pests. The results do not automatically aggregate around certain values, and

there is sufficient space at the higher end of the scale for the pests that pose the highest risk.

In the final model the most influential question was found to be the one about suitability of climate and production conditions ($EST1$), which can potentially nearly double the total risk score. The two questions that can reduce the final score to zero are related to suitability of climate and production practices (if $EST1 = a$, i.e. climate is totally unsuitable) and the host plant area (if $EST2 = a$, i.e. host plants do not exist).

Expert workshops

Three expert workshops were organised to validate the model. One of the workshops was for risk managers at the Finnish Food Safety Authority Evira (11 experts), one for experts on agricultural pests (8 experts) and one for experts on forest pests (7 experts). In these workshops each expert was asked to independently assess the entry, establishment, spread, and impact potential of a set of pests—seven pests in each workshop. Each expert within a workshop assessed the same set of pests, but the set of pests varied between the workshops. The experts had short datasheets for each pest to aid their assessments for the pests they were not familiar with. Following independent assessments, there was a group discussion and a possibility to modify the original assessments. Finally, each participant independently ranked the seven pests according to the level of total risk. In the following text these ranked lists are referred to as expert rankings.

The expert rankings were compared to the simulated median rankings made with the model by an independent expert (J.T.). Equal weight was given to economic (50 %) and environmental/social impacts (50 %). In the following text these ranked lists are referred to as model rankings. The comparison of expert and model rankings was done using Kendall's W which is a coefficient of concordance that can be used to assess whether ranked lists are similar to each other. It varies between 0 (no agreement) and 1 (perfect agreement). Coefficients above 0.9 can be interpreted as unusually strong agreement, coefficients above 0.7 as strong agreement and coefficients above 0.5 as moderate agreement (Schmidt 1997). The expert rankings were compared to each other, and finally the model ranking was included as an

additional expert. This was done separately for each workshop.

The results show a strong or unusually strong agreement between the expert rankings in all the workshops (Table 4). When the model ranking is introduced as an additional expert, Kendall's *W* decreases somewhat for all the workshops. However, the rankings are still in "strong" agreement apart from the forest expert working group where they are in "moderate" agreement.

To analyse whether the model results can be considered as "just another expert", the expert and model rankings were removed one at a time and the resulting change in Kendall's *W* was recorded. The results indicate that in the forest workshop excluding the model ranking results in a larger increase in Kendall's *W* than excluding any of the expert rankings. In the other two workshops, however, there are expert rankings whose removal results in a larger or equal increase than removing the model ranking—implying that the model is more in accordance with the experts as a group than some of the individual experts. This indicates that the model rankings are sufficiently similar and to a large extent in agreement with the expert rankings.

To assess the primary reasons for the discrepancies, the deviation of the expert rankings from the model rankings was studied further. This was done by comparing the placement of individual pests on expert and model rankings. The deviation of 0 indicates that the expert ranked the pest at the same rank as the model, whereas, for instance, +1 indicates that the expert ranked the pest one rank higher than the model. In 29 % of the cases the placement of a pest on the model ranking was identical to the one on expert ranking, and in 61 % of the cases there was at most a deviation of one rank

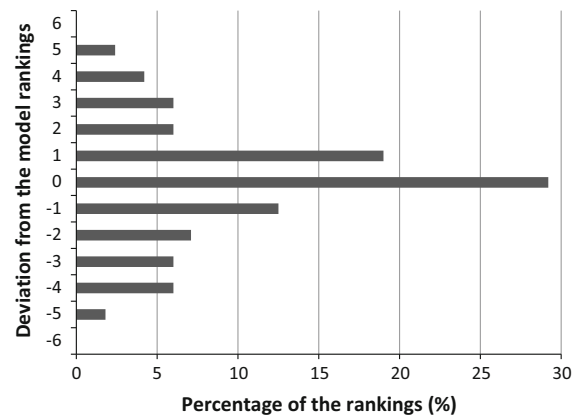


Fig. 1 The deviation between the pest rankings done by the experts in the workshops (expert rankings) and those acquired with the FinnPRIO model (model ranking)

(Fig. 1). Further analysis revealed that there were a few slightly controversial pests that caused most of the disagreements between the expert and the model rankings. These were largely the same pests whose establishment and impact potential was discussed extensively in the workshops.

Model application

To test the ability of the model to differentiate between pest species according to their probability of entry, establishment and impacts, the risk of 95 plant pests was assessed with the model. The assessed pests were selected from a database that included the pests regulated in the European Union's plant health legislation, the pests that EPPO recommends for regulation as quarantine pests or considers to possibly pose a risk (Alert-list species), and pests that are otherwise considered noteworthy for Finland (e.g. Vänninen et al. 2011). First

Table 4 Degree of mutual understanding (Kendall's *W*) regarding the pest rankings between the experts in the workshops (expert rankings) and between the expert rankings and the model ranking

	Degree of mutual understanding	
	Between experts	Between experts and model results
Agricultural workshop (n = 8)	0.925	0.883
Forest workshop (n = 7)	0.703	0.583
Risk manager workshop (n = 11)	0.804	0.750

The results are for ranking seven pests that differed between the workshops

Table 5 Characteristics of the assessed pests according to the type of pest, the legal status of the pest and the plants threatened by the pests

Type of pest	Insects (38 %), fungi (27 %), viruses and viroids (19 %), bacteria and phytoplasma (10 %), nematodes (3 %), mites (3 %)
Legal status	Quarantine pests (80 %), non-quarantine pests (20 %)
Plants primarily under threat	Natural plants (34 %), outdoor ornamentals (32 %), greenhouse ornamentals (28 %), greenhouse vegetables (23 %), berry and fruit plants (22 %), forest and landscaping plants (19 %), potato (11 %), other field plants (11 %), outdoor vegetables (6 %), cereals (3 %), oil plants (1 %)

One pest can threaten several sectors of plant production, and therefore the percentages do not sum up at 100 %

approximately 70 pests were selected on the basis that they represent (1) a wide range of assumed risk; (2) pests threatening different plant production sectors; and 3) different taxonomic groups. The rest of the assessed pests were selected randomly from the database. Some characteristics of the assessed pests are presented in Table 5. The assessments were carried out by one person (J.T.). One assessment took 4–16 h, depending on the pest.

The model was run using Matlab software (Mathworks, Inc.) with 25,000 iterations to ensure stability of the results. The lambda parameter of the PERT distribution, indicating the level of confidence on the most likely estimate, was set at 4, implying a somewhat low confidence on the most likely estimate. The current entry risk management measures were taken into account, i.e. the results shown are produced using question ENT2B instead of ENT2A. Equal weight was given to economic (50 %) and environmental/social impacts (50 %). The simulated section scores, likelihood scores, total risk scores and the resulting ranking of the pests are displayed in Fig. 2, and the respective scores for the feasibility of management in Fig. 3.

Summary statistics on the section and total scores are presented in Table 6. The distribution of the median scores of the entry section is fairly even as is the distribution of the median scores of the establishment section. The impact section is somewhat different: the highest median impact score is fairly low, because in order to get the maximum score from the impact section with an equal weighting of the impacts, the pest needs to have both significant economic and environmental/social impacts.

The rankings based on total risk are fairly stable for most pests. On average, the rank of an individual pest varies 12 ranks in 90 % of the iterations (the difference between the maximum and minimum whiskers in

Fig. 4). However, there are a handful of pests where the variability in rank is very high, indicating a high level of uncertainty. The variability originates from the question related to the suitability of climate and production conditions (EST1)—the question found most influential also in the simulation with random numbers. In the laissez-faire approach the conditions were not considered suitable for these pests. Therefore their minimum score for the establishment and for the total risk is zero, although the maximum answer options selected for these pests yield relatively high total scores.

The results indicate that the model is capable of differentiating between pests based on their probability of entry, establishment, and the likely impacts as well as the total risk. Also the distribution of the scores for the feasibility of management indicates that the model has differentiating capability.

Discussion

Strengths of the FinnPRIO model

The FinnPRIO model was designed such that it can be used to assess all kinds of pests, apart from weeds, in the plant health sector. The model follows the TEASI framework, where the invasion is broken into stages: transport, establishment, abundance, spread and impact (Leung et al. 2012), although in FinnPRIO establishment, abundance and spread are combined into a single section called establishment. The full coverage and separation of the invasion stages allows each stage to be analysed in isolation or in combination with the other stages depending on the specific application. Scoring approaches in general have a reasonably broad coverage of the stages, with 50 % of

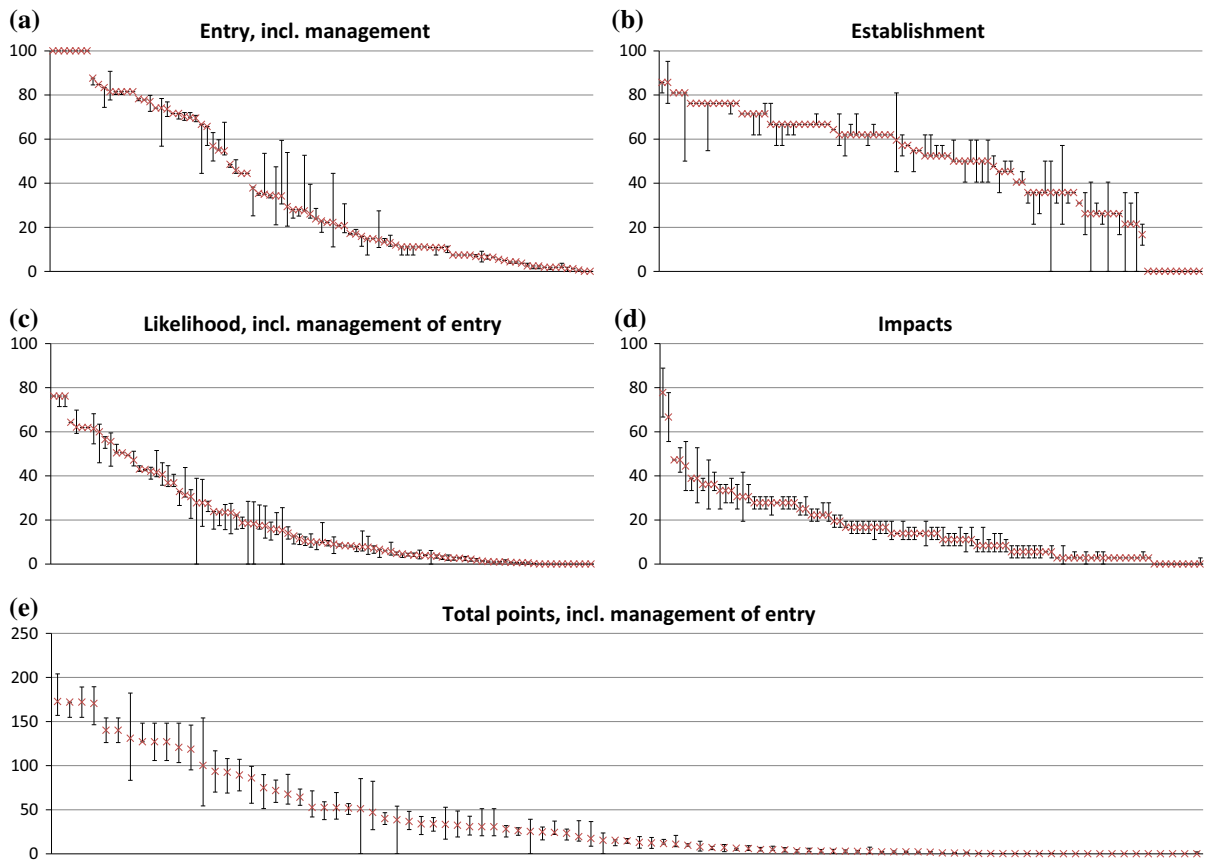


Fig. 2 **a** Entry, **b** establishment, **c** likelihood, **d** impact, and **e** total risk score (y-axis) for 95 pests (x-axis). The cross indicates the simulated median score. The whiskers show the 5th and the 95th percentiles of the distribution of the scores resulting

from the simulation, and therefore indicate the level of uncertainty associated with the assessment. Each section has been individually sorted in a descending order to illustrate the differentiating capability of the model

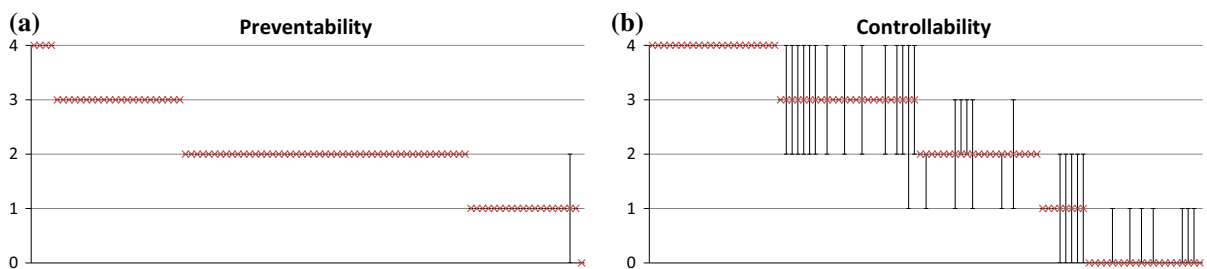


Fig. 3 The scores for **a** preventability and **b** controllability (y-axis) of the 95 assessed pests (x-axis). The cross indicates the simulated median score. The whiskers show the 5th and the 95th percentiles of the distribution of the scores resulting from the

simulation, and therefore indicate the level of uncertainty associated with the assessment. Both scores have been individually sorted in a descending order to illustrate the differentiating capability of the model

them including at least three stages and 18 % including all five (Leung et al. 2012).

In contrast to most existing scoring models for alien species, but in agreement with the most recent models

(e.g. D'hondt et al. 2015; Baker et al. 2014; Moignot and Reynaud 2013), the structure of the FinnPRIO model is fully multiplicative between the sections, i.e. entry, establishment and impact. Multiplicative score

Table 6 The range, mean and standard deviation of the median scores for the 95 assessed pests

	Range	Mean	Standard deviation
Entry	0–100	36	33
Establishment	0–86	49	24
Likelihood	0–76	19	21
Impact	0–78	16	15
Total risk	0–173	36	48
Preventability	0–4	2.1	0.8
Controllability	0–4	2.2	1.4

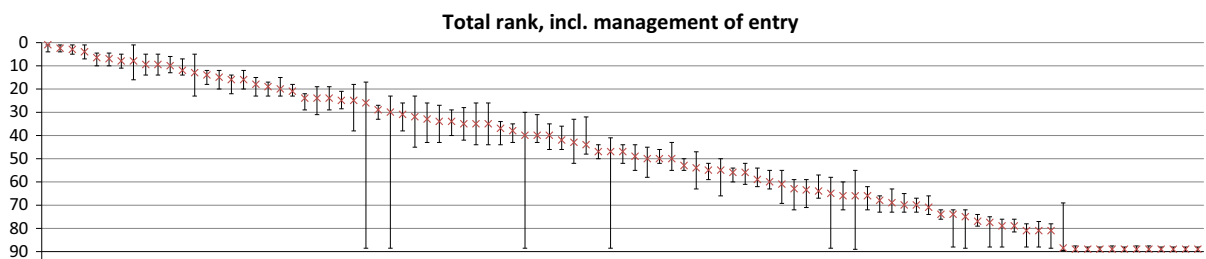
aggregation is recommended by Heikkilä (2011) and Leung et al. (2012) since it describes the total risk more accurately than additive score aggregation. This is because if one of the risk components is judged impossible, the score from the other components should not have any impact on the total risk score (Blackburn et al. 2014a).

In risk ranking there are several types of uncertainty associated with the existence and quality of information as well as regarding the skilfulness of the assessor (Ryan 2006). Such uncertainty is a central component of an informative description of risk, yet it has not been taken into account in about half of the risk ranking systems for alien species (Heikkilä 2011). When uncertainty has been included, it has primarily been used such that some questions can be omitted if the answer is not known. The FinnPRIO model takes uncertainty explicitly into account in each question through most likely, and plausible minimum and maximum answers, and simulates a distribution of the section scores and the total risk score. The assessed pests can be ranked based on, e.g. maximum, minimum, mean or median values of the score distributions or the whole probability distribution can be used to get rankings with a Monte Carlo

simulation, as is demonstrated in this paper. Undertaking a total score simulation and obtaining a range for the rank of the pest is a novel property of the FinnPRIO model.

Invasive species, especially plant pests, can have significant trade and social impacts, yet these are not considered in most of the scoring models (Leung et al. 2012). The FinnPRIO model takes into account environmental, social and economic impacts, including trade impacts. However, potential human health impacts are not included, as these are governed under different frameworks than the plant health context where the model was developed. In the FinnPRIO model, as in some other scoring models (such as D'hondt et al. 2015), it is possible to give different weights to different impact categories. Thus the model can be adjusted to reflect the values of the decision makers.

Risk management measures, such as import bans or special requirements laid down in plant health legislation can mitigate pest risk considerably (Kehlenbeck 1996). Taking into account the effect of current control measures and the feasibility of further measures could therefore make the results of scoring models more informative. In the FinnPRIO model the current management measures are considered in the entry section, where the potential of the pest to be transported via a given pathway is assessed both with and without the requirements set in the relevant legislation. This dual assessment allows ranking of pests with regard to both the unmitigated and mitigated entry risk, which also makes the importance of the current legislation detectable. The feasibility of further risk management measures is assessed in a separate management section, where both preventability and controllability of the pest are assessed. This feature is lacking in most other alien species scoring

**Fig. 4** Ranking (y-axis) for the 95 pests (x-axis) according to their total risk score. The cross indicates the simulated median rank. The whiskers show the 5th and the 95th percentiles of the

distribution of the ranks resulting from the simulation, and therefore indicate the level of uncertainty associated with the assessment

models, since about 40 % of them do not cover either preventability or controllability and very few include both (Heikkilä 2011). Even some of the more advanced recent scoring models (such as *Harmonia*⁺ and *Pandora*⁺, D'hondt et al. 2015) exclude management feasibility.

To facilitate consistency between assessments of different pests, and assessments done by different assessors it is important that there are clear verbal descriptions of each question and answer option (Leung et al. 2012). The Australian WRA, which is probably the most widely used scoring model, has a guide that describes the response options in detail (Gordon et al. 2010), as do some of the recent model developments (e.g. D'hondt et al. 2015). Similarly, a guide has been developed for the FinnPRIO model (available as supplementary material). It has question-specific instructions on which factors should and should not be considered when answering the questions, as well as examples of cases that would correspond to the different answer options. However, expert models always have some scope for expert opinion, and therefore a fully unambiguous guidance is not realistic.

A scoring model needs to be validated to ensure that it does what it is supposed to be doing. There are several ways to do validation, yet many of the alien species scoring models have not been validated or validation has not been reported (Heikkilä 2011). The most common validation has been to test the model with species whose pest status is known, and assess whether the model produces similar results. The FinnPRIO model has been validated with simulation analysis, expert workshops and by applying it to a considerable number of pest species.

The FinnPRIO model and its question-specific guidance are specifically designed for ranking the risks of invasive pests for Finland. Yet, with a few modifications the model may be applicable in other countries too. Especially the answer options of the questions related to volume of trade (ENT3), host plant area (EST2), and direct economic impacts (IMP1) should be rescaled. If applied outside the European Union, the question related to the presence of the pest in the European Union (MAN2) needs to be revised. Still, some of the rating criteria, such as the impacts on culturally important plants (IMP4.3), are not necessarily considered important in other

countries, while the FinnPRIO model might miss some criteria considered relevant elsewhere.

Challenges in building scoring models for invasive plant pests

Building a scoring model that describes a complex process, such as pest invasion and its consequences, enables comparison of a myriad of cases and is simple and quick to use, is challenging. The model is bound to be a coarse simplification and a number of shortcomings have to be tolerated.

Deciding how to aggregate the factors that need to be included in the model into questions of a scoring model is not straightforward. At one extreme each question deals with just one well defined aspect of the process, for instance the effect of plant protection measures on the pest prevalence and abundance in the country of origin. Such an approach is transparent and its outcome is likely to vary very little between different assessors. However, if the number of questions is high, defining how the question scores should be aggregated into section scores in a manner that accurately describes all imaginable cases can become impossible. At the other extreme an estimate for the whole section (i.e. entry, establishment or impact) is asked in just one question and the decision on how to weigh the different factors affecting the section score is left to the assessor. In such a model special cases are easier to deal with, but the assessment is not as transparent and variation between different assessors may be very large. The number of questions in the FinnPRIO model is actually close to the median number of questions (17) in existing scoring models (Heikkilä 2011).

Similar issues arise when the number of answer options in the questions is either very high or very low. A large number of answer options makes consistency between different assessors more difficult, whereas too few answer options make it more difficult to differentiate between the pests and to express the uncertainty of the assessment. Furthermore, it has been shown that if the number of alternatives gets close to ten the results get biased downwards compared to five and seven point scales (Dawes 2008; Makowsky and Mittinty 2010). It has also been argued that in such cases the answer alternative cannot be clearly defined (MacLeod and Baker 2003).

The interaction of the spread section with the impact section needs to be designed so that the combination of the two terms is meaningful. The expected spatial extent of the impact should be incorporated in either the spread or the impact section, but not both, if the two sections are multiplied to get the total risk score. In the FinnPRIO model the impact section measures the expected impact of the pest in the case in which it has spread throughout its potential area of establishment.

The treatment of pests that either need a vector or are themselves vectors for other pests in a scoring model should be defined clearly. The vector and the pest that it transports can be assessed either separately or together (similar to pathogen-product combinations in food safety, e.g. Mataragas et al. 2008), but in both cases their effect on each other's probability of entry, establishment, spread and impact should be taken into account consistently (see, e.g. Baker et al. 2014). In the FinnPRIO model each pest is assessed separately, and the question-specific manner of assessing vectors and pests which need a vector is specified in the guidance.

Assessing the impacts of pests that have multiple host species using semi-quantitative scores is also not straightforward. With a quantitative economic estimate of the total potential damage, this would present no problem. But, on a semi-quantitative scale, pests that have several hosts become problematic if some questions (say, spread potential) are answered in relation to one host species, and others (say, potential economic impacts) in relation to another host species. Combining these answers (to conclude fast spread and severe impact) may be fallacious. Theoretically it would make sense to answer all the questions separately in relation to all the host species, but often this is not feasible given the time constraints for the assessment. It is therefore necessary to give clear guidance on how to treat such cases in scoring models. In the FinnPRIO model all host plants are taken into account in all the questions, and the question-specific manner of doing that is specified in the guidance.

Also estimation of economic, environmental and social impacts on an equivalent semi-quantitative scale presents a challenge. In a quick scoring analysis monetisation of all impacts cannot be done, and therefore "significant" environmental impacts are unlikely to correspond exactly or even approximately to "significant" economic or social impacts (e.g.

Sandvik et al. 2013). So far a clear solution to this problem does not exist, although there are some recent attempts to address the issue.

Taking into account current risk mitigation measures in defining risk scores may be either desirable or undesirable, depending on what the scoring is used for. Exclusion of the mitigation measures is required if the aim is to compare pest species as such, for example to demonstrate the importance of risk mitigation measures. On the other hand, inclusion of risk management measures is the more desirable alternative if the aim is for example to identify pest species that deserve further regulation or surveillance. In the UK plant health risk register assessments are made with and without the current risk management measures (Baker et al. 2014). Similarly, in the FinnPRIO model the entry score can be assessed both with and without current entry risk management measures. However, we recognise that the assessments where the mitigation measures are excluded are likely to be less reliable, mainly because predicting trade flows in the absence of current import regulations is mostly guesswork.

Using scoring models for policy recommendation

The objective of scoring models is usually to aid decision making related to risk management, and consequently scoring models are relatively commonly applied for policy support (Leung et al. 2012). Translating scoring results into policy recommendations is not always straightforward, and results that are not communicated properly can be misinterpreted.

The total risk score is an appealing measure for risk communication as it combines all the components of risk into one figure. However, it is seldom sufficient for planning risk management measures. Similar total risk scores can be composed of very different section scores, for instance a high entry score and a low impact score or vice versa. When deciding on which species risk management resources should be targeted, such differences become important. For the same reason, the section scores cannot be considered in isolation either. A species with a high section score for, say, entry, may become irrelevant from management perspective if its section score for establishment or impact is insignificant. It is therefore useful to study the scores of the different sections both individually and in combination when identifying

species for management. Alternatively, priority species can be identified systematically using the multiattribute frontier concept to combine the scores of the different sections (e.g. Yemshanov et al. 2013b).

Taking into account the manageability of the pest is crucial when designing risk management measures. It may be, for instance, especially desirable to impose strong import measures to prevent entry of a species whose post-entry spread within the country is difficult or impossible to manage. Obviously, it is also useful to recognise the risks that cannot be managed: for example, if a pest is likely to spread to the country naturally, any import restrictions are obsolete.

Uncertainty is a key factor in risk assessment, and it should also be included in the output of the assessments. Depending on the perspective of the decision makers they may wish to look at the mean or the median risk score, the maximum score (precautionary approach) or the minimum score (laissez-faire approach). Although the decision makers often prefer simple answers, presenting the uncertainty adds relevant information to support sound science-based decision-making.

Conclusions

An output of a pest scoring model is a simplified uncertain depiction of reality, the reliability of which depends on the quality of the input data, the skilfulness of the expert using the model, and the way the model processes the input data into outputs. Despite these challenging characteristics of scoring models, the FinnPRIO model is judged to perform adequately for the job that it was designed to do: to quickly assess the risk presented by a plant pest, and to provide support for decision making.

The recent recommendations for scoring models (Heikkilä 2011; Leung et al. 2012) were taken into account and implemented in the development of the FinnPRIO model, enabling the testing of the feasibility and functionality of the recommendations in practice. Later on the FinnPRIO model can be used when scrutinising the value added of these recommendations by comparing models with different attributes. Based on an objective comparison of different approaches, future development of pest risk scoring models might concentrate on fine-tuning some of the

most prominent models. Taking into account the limitations in data and resources available for assessment, rapid risk assessment approaches will likely continue to be needed.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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